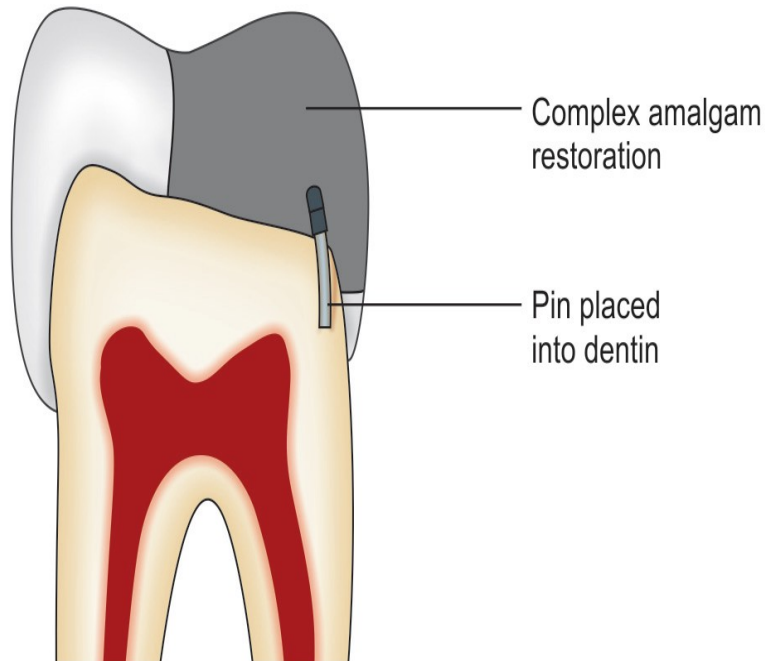


PIN RETAINED RESTORATION



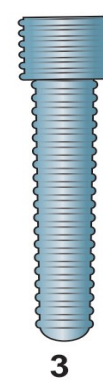
Minuta



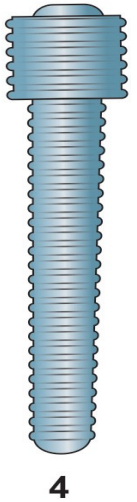
Minikin



Minim



Regular



INTRODUCTION

When Does pin retained restoration?

- **W**hen the **preparation size is very big** due to caries or other reason and
- **W**hen the **remaining tooth structure is very less**, it becomes difficult to achieve optimal resistance and retention form.
- **W**hen **dentin retentive features** (lock and slots) prepared in the dentin are **insufficient** to provide desired retention, pin supported restorations are used.

What is PRR ???

Definition

- It is defined as any restoration which **requires the placement of pin/pins** in the dentin in order **to provide sufficient retention and resistance form** to the restoration.

Purposes

1. **Pins provides;** retention and resistance form:

- Retain the restoration in the prepared tooth structure together

2. **Pins provides;** support and strength.

- Reinforcement the tooth by cross-splinting of weakened cusps.

INDICATIONS OF PRAR

1. Grossly mutilated teeth:

- **Large restorations** may be fractured or may be lost due to occlusal loading.
- **In cusplless vital teeth** use of pins will provide retention and resistance.

2. Extended preparations:

- Use of pin provides added strength **when the margins of the restoration are not surrounded by enough tooth structure.**

3. Cores for full coverage restoration:

- By pin-supported amalgam, core can be placed in less time and expense.
- The retention and seal produced by the pins preserve the tooth during the **extra coronal restoration.**

Cont.....

4. Extensive class V restorations:

- The **success rate of extensive class V restoration is increased by the use of pins.**
- Pin placement also tends to reinforce the restoration.

5. Time period and Economic factors:

- When there is **no chance for second appointment or an economic problem** regarding cast gold restoration, pin-retained amalgam restoration can be the restoration of choice.

CONTRAINDICATIONS OF PRAR

- When patient has occlusal problems.
- When esthetics is concerned.
- When **direct restoration is not possible** because of functional or anatomical considerations.

ADVANTAGES Of PRR

1. Conservation of tooth material:

- Where enamel is supported by **sufficient amount of healthy dentin** and the **area is not exposed to heavy masticatory forces**, moderate amount of undercut can be given. Therefore, the preparation is **more conservative than** slot and lock preparation or tooth preparation for cast restoration.

2. Number of appointments:

- **Only one appointment** is required for PRR , whereas for cast restoration, at least two appointments are required.

Cont.....

3. Resistance and retention form:

- Use of pin **increases the resistance and retention** of the restoration and requires minimum cutting of healthy tissues.

4. Economics

- Pin restoration is relatively **inexpensive**, does not require laboratory facility and can be completed in single sitting compared **to cast restoration**.

DISADVANTAGES

1. In pulpectomized and pulpotomized teeth:

- In such teeth **elasticity of dentin gradually reduces**. Therefore, PRR are contraindicated because the **chances of fracture of dentin increase due to its dehydration**.

2. Dentinal microfractures:

- When minimal dentin is present, fracture lines in dentin or internal stress in dentin may be created while **drilling pin holes or during pin placement**.

3. Microleakage:

- Microleakage may occur around the pins if **their ends appear on or near to the surface of the**

Cont.....

4. Strength of amalgam:

- Pins decrease the tensile and transverse strength of amalgam while compressive strength is not increased.

5. Perforations:

- On misdirection of the bur or pin there is a risk of perforating into the pulp or the external tooth surface.

6. Tooth anatomy:

- In some cases it is not possible to achieve proper contours and occlusal contacts with PRRs.

Retentive pin

Introduction

- In 1958, Dr Miles Markley introduced **stainless steel pins**(SSP) to provide retention and resistance form.
- Pins are **available in different shapes, sizes and materials** such as stainless steel, platinum-palladium, platinum iridium, plastic, aluminum and acrylic.

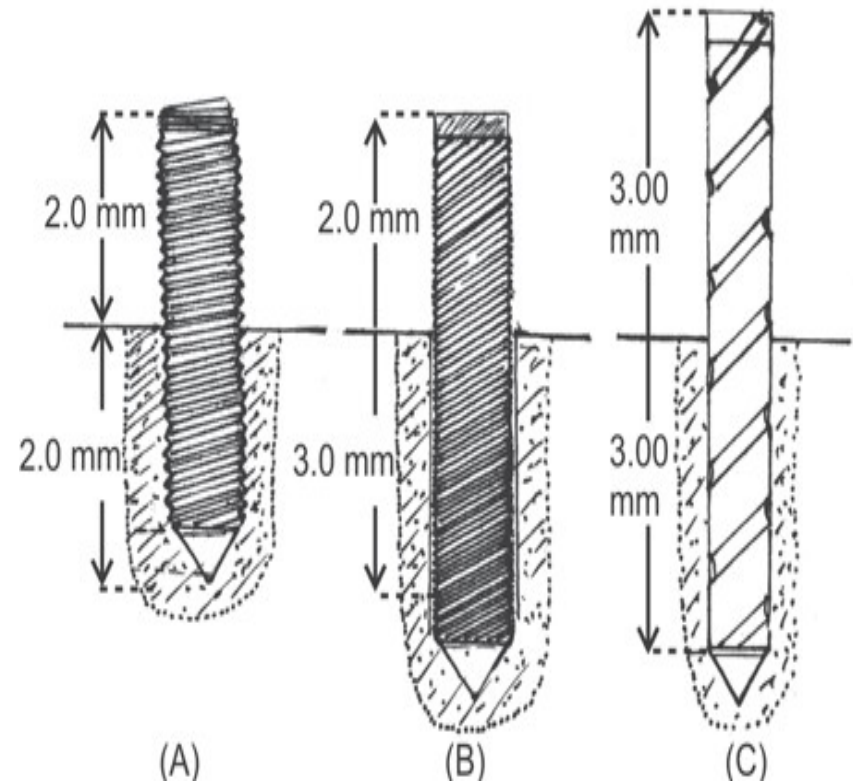
Types of pins

Pins can be classified as:

❖ 1. Direct pins/non-parallel pins

- A. Self-threading pins
- B. Cemented pins
- C. Friction locked pins

2. Indirect pins/parallel pin



1. Direct pins

- These pins are also called as the **nonparallel pins** because they can be inserted **directly into the tooth structure** and need not be parallel.
- Are generally made of **stainless steel** and it can also be made from other materials like silver, titanium, stainless steel with gold plating, etc.
- Are inserted into dentin, after this restoration is placed directly over them.

2. Indirect pins

- These pins are also known as the **parallel pins** because these pins are **placed parallel to each other and also to the path of insertion** of the restoration.
- Are made **smaller in size when compared to their pinholes** and they constitute an **integral part of a cast restoration**.
- **Two types of pins are used in the parallel pin technique:**
 - a. **Cast gold pins:** These pins have a smooth surface.
 - b. **Wrought precious metal pins:** Surface of these pins has been roughened by means of threaded or knurled patterns.

A. Self-threading pin:

Indications

- In vital teeth
- When maximum retention is desired
- When sufficient amount of dentin is available to surround the pins.

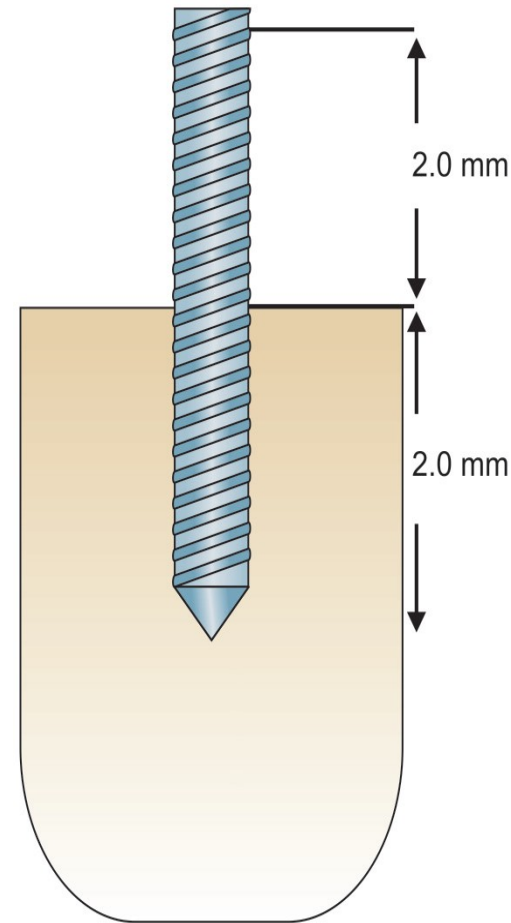


Figure 5.1: Self-threading pin

Facts: PH<PD

- They were introduced by **Going in 1966.**
- For these pins the **pinholes (PH)** are slightly smaller (0.038 to 0.1mm) in diameter than the **pin diameter (PD).**
- Self-threading pins are 0.0015" to 0.004" larger than their pinholes.
- The pins are retained due to:
 1. the **mechanical grasp of the threads into the dentin.**
 2. the elastic property of the dentin allows insertion of a threaded pin into a hole of a smaller diameter.
- These are available in stainless steel or titanium but can be gold plated to increase their passivity.

Cont.....

Advantages

- Ease of insertion.
- Superior retention.
- Require less depth for placement(conservative)
- Require no cementing medium.

Cont.....

Disadvantages

- **Pulpal stress is maximum-** **when** the pin is inserted perpendicular to the pulp
- **Generate great stresses-** in dentin in form of craze or crack lines.
- Pins may **fail to seat completely**
- **Strip the sides of the dentin** resulting in a loose fit - If pin is forced into the pinhole.
- **Microleakage is higher** than the cemented pins if the overlying restoration leaks.
- Pins may need to be **bent, cut or contoured**

B. Cemented pins:

Indications

- In cases where **least stresses or crazing is desired.**
e.g. endodontically treated teeth
- When **bulk of dentin to hold the pin is less.**
- When **dentin has lost its elasticity** because of dehydration or sclerosis
- When pin has to be **placed near DEJ.**

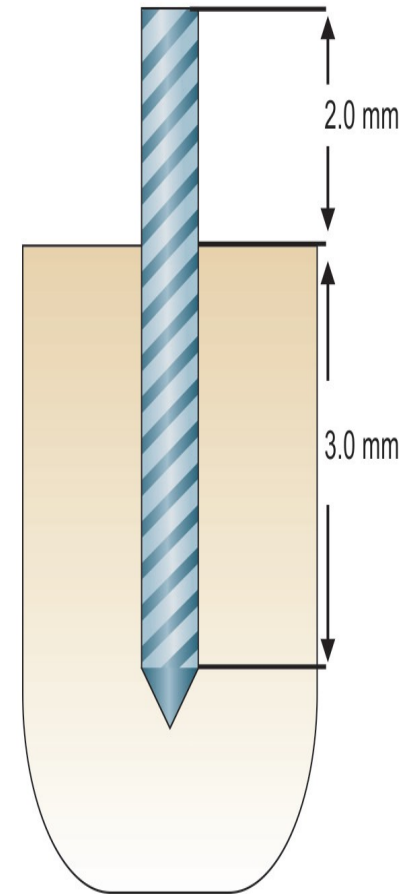


Figure 5.2: Cemented pin

Facts: PH>PD

- Introduced by Dr. Merklay in 1958 for obtaining greater retention in large amalgam restorations.
- Are available in **various diameters ranging** from **0.018" to 0.030"** with the corresponding **pinholes** of 0.020 to 0.032 inches.
- In general for these pins, **the prepared pinholes should be larger than the diameter of the pin by 0.025 to 0.05 mm.** This difference in diameter provides space for cementing media.
- The depth of hole in dentin should be 3 to 4 mm.

Cont.....

Advantages

- Cemented pins are **0.001 to 0.002 inches smaller** than their pinholes. Thus they can be seated to the full depth of the hole.
- Since they are **passively retained in the dentin**, they produce no stress on the surrounding dentin
- Less microleakage
 - Cement present between the pin and the tooth decreases the chances of microleakage

Cont.....

Disadvantages

- Provide less retention.
- It is **difficult to locate the pinhole after cement has been placed** in it for cementation
- A **poorly cemented pin can be dislodged** while inserting filling material.
- **Requires more time for the mixing** and hardening of the cement.

Techniques of pin placement

- **Dry the prepared pin channels** with endodontic paper points.
- **Place the cement mix** into the pin channel with a root canal file or/an explorer or a lentulospiral running at slow speed of 1000 rpm.
- **Hold the pin in the forceps and coat with the cement,**
- **Insert** the pin into the hole.
- **Hold the pin in its position** till it sets.
- **Remove excess cement** with an explorer.

C. Friction-locked pins:

Indications

- Vital teeth
- Periodontally sound teeth
- When **direct access is possible** so that the tapping force can be applied parallel to the long axis of the pin
- When **sufficient amount of dentin is available** to surround the pin.

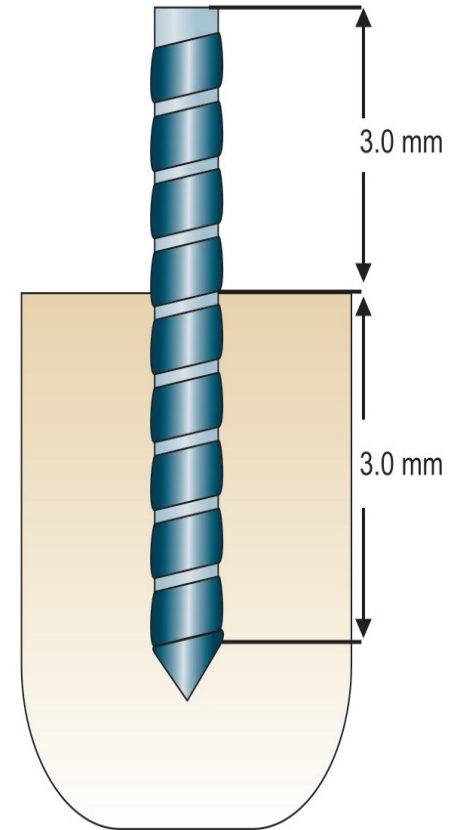


Figure 5.3:
Friction locked pin

Facts: PH<PD

- Introduced by Goldstein in 1966.
- The diameter of the **pinhole is smaller than the diameter of the pin by 0.025 mm**. These pins are 0.001" larger than their pinholes.
- Pin retained, hence utilize the elasticity of dentin for retention.
- By gentle tapping they are placed into dentin and are retained due to **resiliency of dentin**.
- They are 2 to 3 times more retentive than the cemented pins.

Cont.....

Advantages

- Cementing media not required.
- Pins attain stability from the moment they are inserted.

Disadvantages

- The length of the **pin cannot be adjusted outside** after insertion.
- Bending or contouring of the pin after it has been inserted into the pinhole results in further stresses
- Sometimes the **pins do not reach full depth of the channel** because of gouging

Techniques of pin placement

- **Cut the pin to the desired length** before insertion since pin cannot be inserted for try-in and adjustment.
- **Insert the pin into a pin setter** and carry to the pinhole.
- **Apply force with mallet parallel to the long axis** of the pin until pin reaches the desired length
- **Remove excess length** with a small round bur
- **Bend the pin at a desirable angle** with contouring pliers, if desired.

SUMMARY

Type of pin	Diameter of pinhole	Pin depth in dentin	Pin depth in amalgam
Cemented pin	0.025–0.05 mm larger than diameter of pin	3–4 mm	2 mm
Friction locked pin	0.025 mm smaller than diameter of pin	3 mm	2 mm
Self-threading pin	0.038–0.1 mm (0.015–0.004 inches) smaller than diameter of pin	2 mm	2 mm

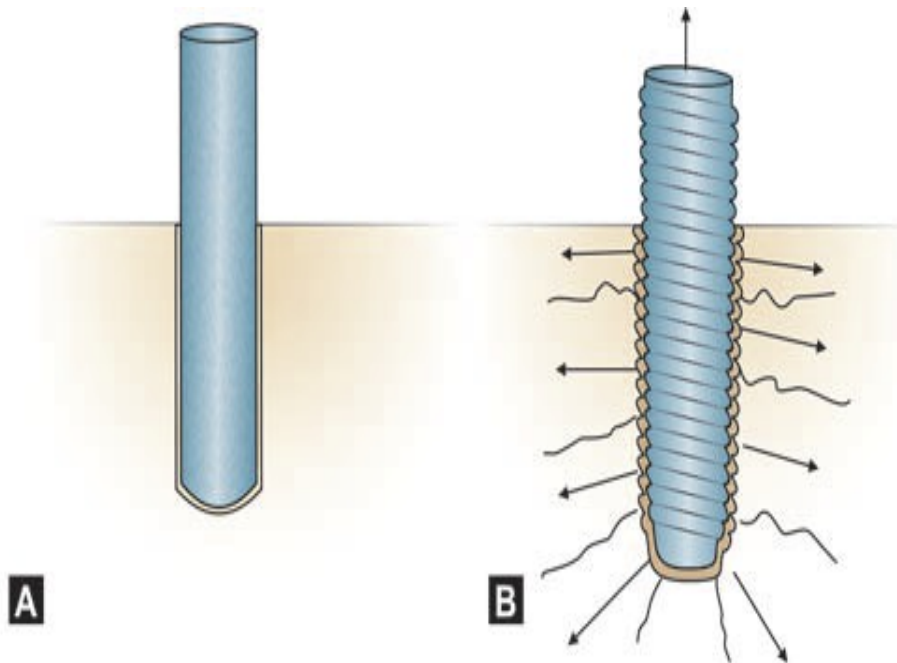
PRINCIPLES AND TECHNIQUES OF PIN PLACEMENT

1. Pin size

- In general, increase in diameter of pin offers more retention but large sized pins can result in more stresses in dentin. Pins are available in four color coded sizes:

<u>Name</u>	<u>Color code</u>	<u>Pin diameter(size)</u>
A. Minuta	pink	0.38 mm (0.015 inch)
B. Minikin	red	0.48 mm (0.019 inch)
C. Minim	silver	0.61 mm (0.024 inch)

Cont.....



Figs 5.5 A and B: If pin size is greater than pin channel it can cause stresses in the dentin

A. Pin diameter less than pin channel= **microleakage**.

B. Showing stresses in dentin when pin diameter is more than pin

Selection of pin

❖ Selection of pin size depends upon following factors:

- Amount of dentin present.
- Amount of retention required.

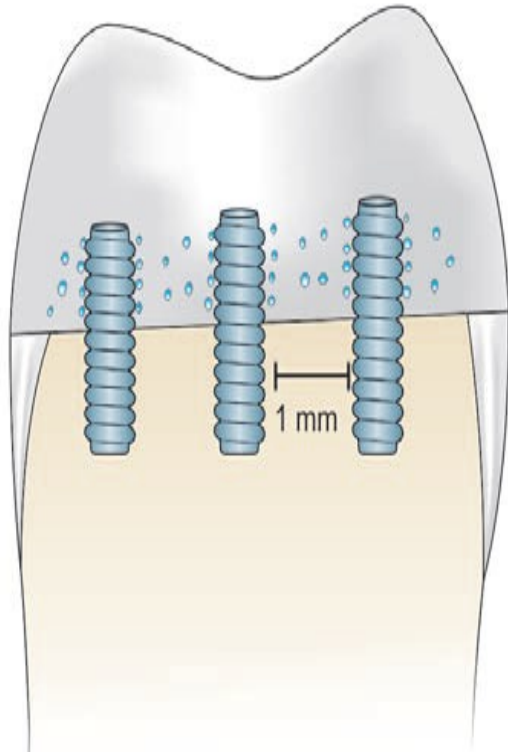
NOTE

- For most of the posterior restorations, **minikin or minim size** of pins are used because they provide maximum retention without causing crazing in dentin.

2. Number of Pins

- General rule: one pin per missing cusp and one pin for each missing axial line angle.
- **Unnecessary use of pins** should be avoided as they can cause following (**Fig.5.6**):
 - Generate stress in the tooth resulting in fracture of tooth.
 - Voids in restoration.
 - Decrease the amount of available inter-pin dentin.
 - Decrease the strength of amalgam restoration.

Cont.....



Factor affecting choice for number of pins used:

- Amount of missing tooth structure.
- Amount of retention required.
- Amount of dentin present.
- Size and type of pin.

Fig. 5.6: Unnecessary use of pins can cause stresses and voids in restoration

3. Interpin Distance

- When more than two pins are to be used, inter-pin distance should be such that;
 1. It prevents concentration of stresses in dentin and
 2. It allows space for insertion of restorative material between pins.

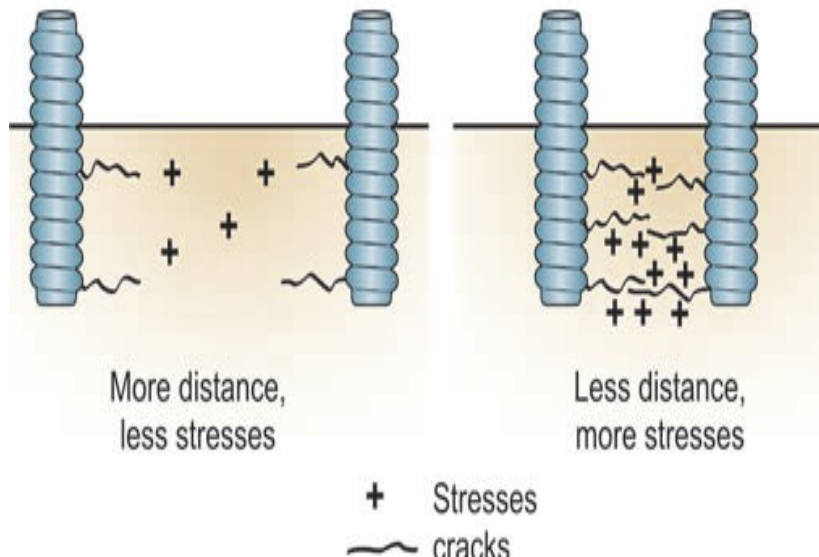
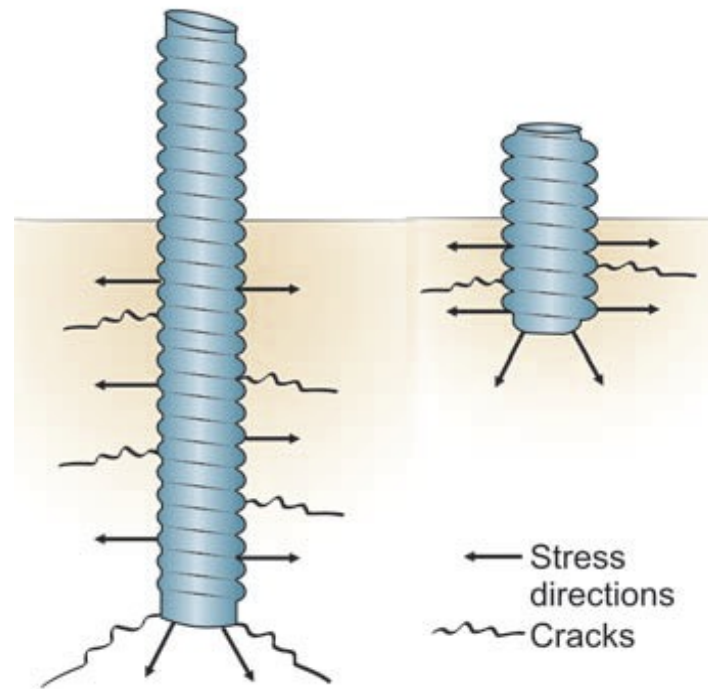


Fig.5.7: Lesser is the interpin distance, more are the stresses generated

4. Length of Pin into Dentin and Amalgam

- In general, **pin extension of 2 mm into dentin and amalgam provide maximum required retention.**
- Pin extension **more than 2 mm is avoided** so as to preserve the strength of dentin and the restoration.

Fig. 5.8: More is the pin in the Restoration or dentin, more are the stresses



5. Location of Pin Placement

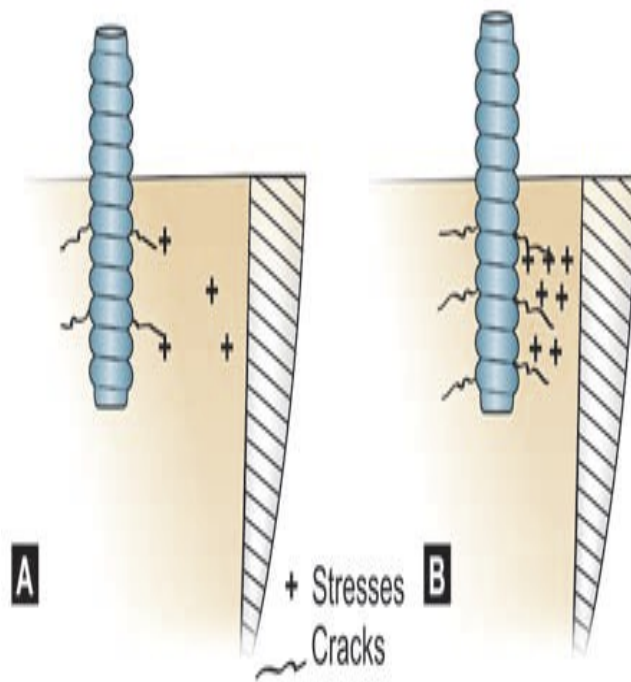
- **Following factors should be considered while selecting location of pins:**
- **K**nowledge of normal pulp anatomy to avoid pulpal exposure or external tooth perforations.
- Avoid placing the pins directly under occlusal loads as this may weaken the amalgam.
- Pinhole should be at least 0.5 mm inside the dentin to reduce the chances for crazing of the tooth structure.
- Pinholes should be located on a flat surface.

Cont.....

- There should be atleast 1mm of dentin between pulp and the pin to avoid pulpal damage.
- Intermittent radiographic monitoring should be done constantly, while preparing and placing the pins.
- When more than two pinholes are planned, they should be placed at different levels to prevent the stresses in same transverse plane of the tooth.
- If two or more pins are to be placed, inter-pin distance should be such that it results in lower levels of stresses in dentin and

Cont.....

- There should be at least 1mm of sound dentin around circumference of the pin (**Figs 5.11A and B**).



Figs 5.11A and B:

(A) There should be at least 1 mm of sound dentin around pin circumference

(B) Lesser dentin present around pin results in more stresses

IDEAL PLACEMENT

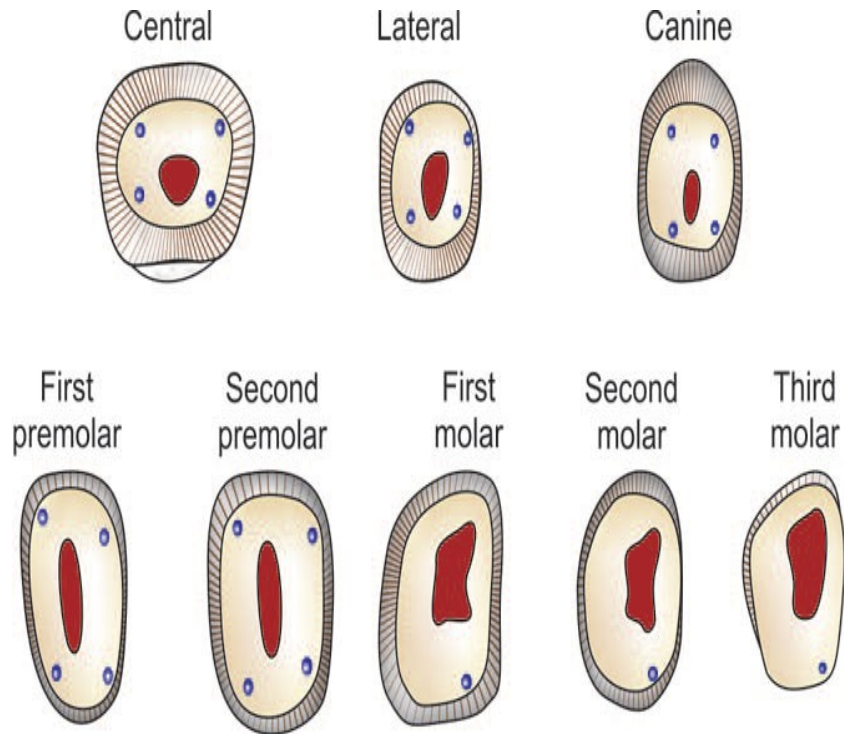


Fig. 5.9: Ideal site for pin placement in maxillary teeth

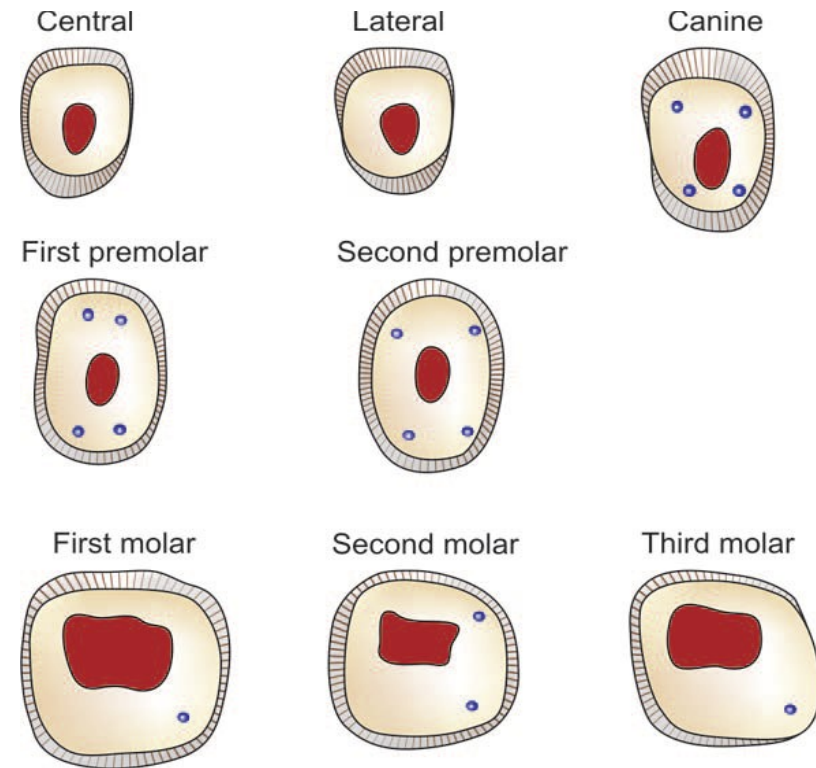
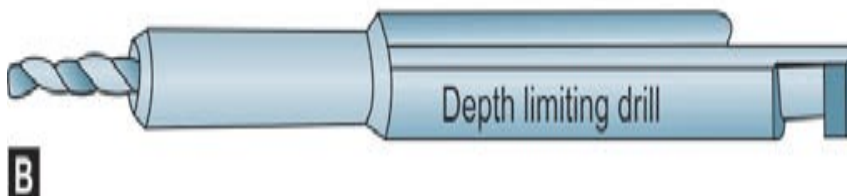


Fig. 5.10: Ideal site for pin placement in mandibular teeth

6. Preparation of Pinhole

- Pinholes are prepared using twist drills. Commonly used drill for pinhole preparation is **Kodex drill**.
- The drill is made of a high-speed steel that is swaged into aluminium shank. Drill performs cutting when rotated clockwise at slow speed. Suggested speed for drilling is 300-500 rpm to 1000



Figs 5.12 A and B: Twist drills

(A) Standard drill

(B) Depth limiting drill

Cont.....

- Direct the drill towards the **desired location of pin placement**.
- Drill should be kept continuously moving **only in one axis and one direction** from the time of insertion till removal to avoid fracture of the drill in the pinhole and over cutting.

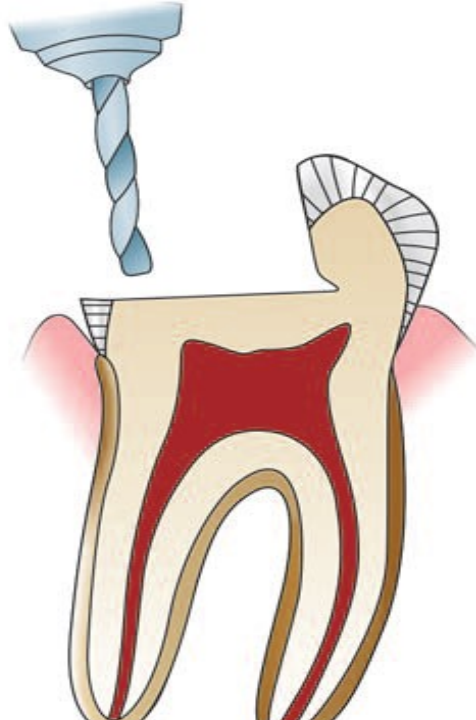


Fig. 5.12: Direct the drill towards the desired location of pin placement

Cont.....

- While cutting dentin, apply slight pressure.
- During drilling, AVOID
 - **Slanting of the handpiece,** or
 - **Allowing the drill to rotate more** at the bottom of the pinhole as this may results in a large hole.

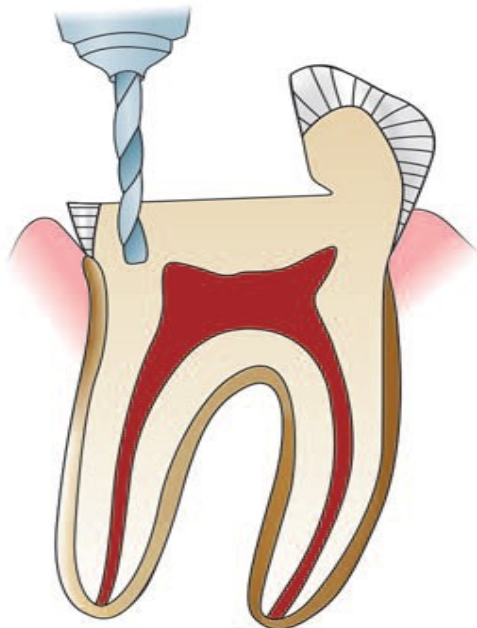


Fig. 5.13: Preparation of pinhole

FACTORS AFFECTING RETENTION OF PINS IN TOOTH STRUCTURE

- Several factors are known to control the retention of pins in tooth structure. These include:

1. Pin Diameter

- Within limits, **retention is directly proportional to diameter of pin.** For example, retention doubles when the diameter of the pin increases from 0.0155" to 0.0190".
- However, overzealous increase in diameter may decrease amount of dentin and thus weaken the tool.

2. Pin Number

- Within limits, **increasing the number of pins increases the retention in dentin.**

3. Pin Type

- Cemented pins are the least retentive.
- Friction locked pins show intermediate retention and threaded pins are the most retentive. Friction locked and threaded pins are retained by the elasticity of dentin and this accounts for their higher retention.
- Order of retention offered by different pin systems is: **threaded pins > friction locked pins > cemented pins.**
- Within the cemented pin type, **serrated pins are more retentive than smooth surface pins.**

4. Pin Depth in Dentin

5. Cementing Agents

- **Zinc phosphate cement** is more retentive than PCC and ZnOE cements.
- Glass ionomer cement is also more retentive. However, varnish reduces the retentive ability of the cemented pins.

6. Association between Pin Channel and Pin Circumference

- Poor quality in the manufacture of pins can result in problems like mismatch between the pin and drill diameters, variations in the inside diameter and thread shapes of the pins, etc.

7. Type of Dentin

FACTORS AFFECTING RETENTION OF PINS IN RESTORATIVE MATERIAL

❖ The following factors affect the retention of pins in restoration material:

1. Pin Length

- Within limits, **increasing the length of the pin in restorative material increases retention.**

2. Pin Number

- Within limits, **increasing the number of pins increases retention of the restoration.**

NOTE. Excessive increase in number of pins makes condensation of restorative

material difficult and decreases the overall strength.

3. Pin Diameter

Cont.....

4. Pin Orientation

- Pins placed in nonparallel increase retention.

5. Pin Shape

- Retention cleats and square or pear shaped heads on the pins improve retention of pin.

6. Interpin Distance

- Placing pins close to each other (minimum interpin distance 2 mm) increases retention. At distances lesser than 2 mm pin retention is reduced because of the less amount of material present in between the pins and increase in residual stresses in dentin.

Cont.....

7. Pin Restoration Interphases

- An ideal interphases between the pin and material is one which is not interrupted.

8. Surface Characteristics

- Number of serrations present on the pin surface affect its retention.

9. Bulk of Material Surrounding the Pin

- Pin retention is directly proportional to bulk of material surrounding the pin. Retention is almost lost when the material around the pin is less than half the diameter of the pin.

PINS AND STRESSES

1. Pins, Stresses and Tooth

- Stresses are seen maximum with use of friction locked and threaded pins in dentin. Stresses are developed since
- pins are inserted into channels 0.001-0.004 smaller than the diameter of the pins. If the stresses exceed dentin's plastic limit, craze lines or cracks are seen. These fracture lines can cause pulpal involvement.

NOTE

- Threaded pins show only apical stresses whereas friction locked pins act as wedges which generate lateral stresses, cracked tooth syndrome, gross fractures, loose restorations, etc.
- Cemented pins are shown to induce the least stresses, threaded pins induce intermediate stresses and friction locked pins induce the maximum stresses.
- Stress tolerance of different types of dentin in a decreasing order is:
secondary dentin > sclerosed dentin > tertiary dentin > calcific barrier.
- Cemented pins are therefore, the only preferred pins in endodontically treated teeth with non-vital dentin.

Cont.....

2. Pins, Stresses and Restorative Material

- Pins did not strengthen or reinforce a restoration but aid in retention of restoration. Neither compressive strength nor transverse/tensile strength of amalgam is improved because of stress concentrations around the pins and cleavage planes set up in the restoration by the arrangement of pins.
- Pins are likely to reduce the strength of amalgam because of absence of any chemical union between the pin and restorative material at the interface.

FAILURE OF PIN RETAINED RESTORATIONS

- The pin retained restoration may fail due to one of the following reasons.

A. Within the Restoration

- Restoration may fracture, may be because of improper condensation or improper trituration.

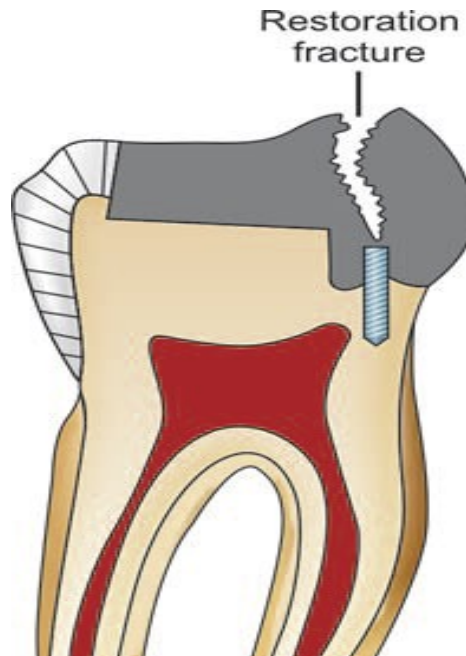


Fig. 5.14: Restoration fracture

B. Within Pin

- Pin fracture may occur because of improper technique during pin placement (**Fig.5.15**).
- Broken pins can occur in following conditions:
 - During bending or if turned more than required in the pinhole.
 - Excessive force is applied while its placed
 - Pin is rotated despite being fully seated in pinhole.

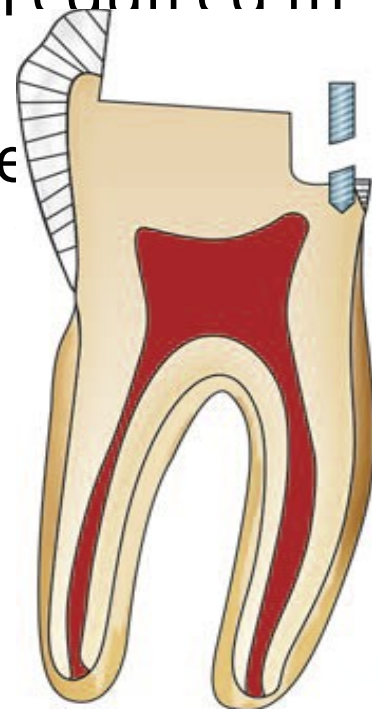


Fig. 5.15: Pin fracture

C. At Pin Restoration Interface

- Restoration may pull away from the restoration because of corrosion products at interface
(Fig.5.16

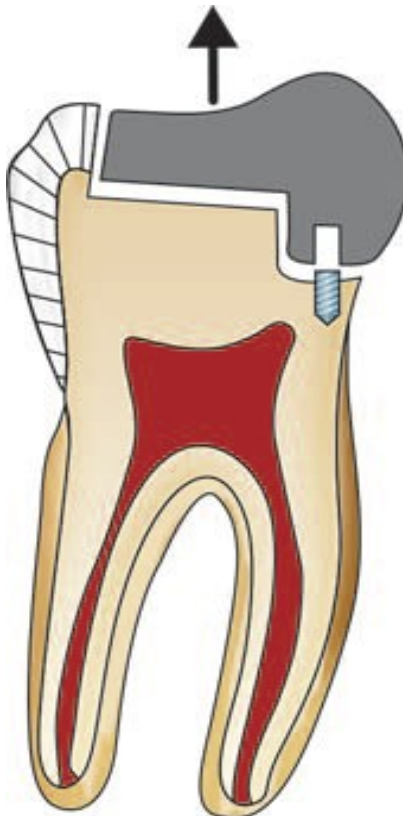


Fig. 5.16: Restoration pulled away from the p

D. At Pin Tooth Interface

- Pin may separate along with restoration because of improper pin tooth joint (**Fig.5.17**).

E. Within Tooth

- Dentinal fracture (**Fig.5.18**) can occur because of concentration of internal stresses because of selection of pin type.

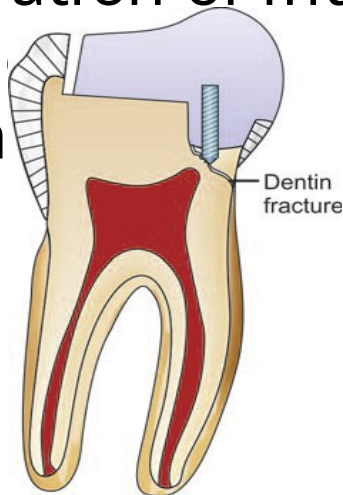


Fig.5.18: Dentin fracture

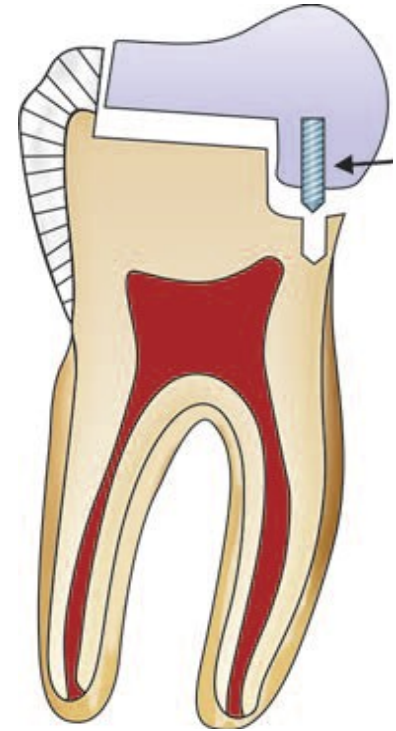


Fig. 5.17: Separation of pin along restoration

F. Broken Drills

- Drill can break in following conditions:
 - Drill is stopped before being removed from the pinhole
 - Stresses are applied laterally during drilling.
 - Dull drills are used.
 - Drill is stopped while entering or exiting from the tooth.

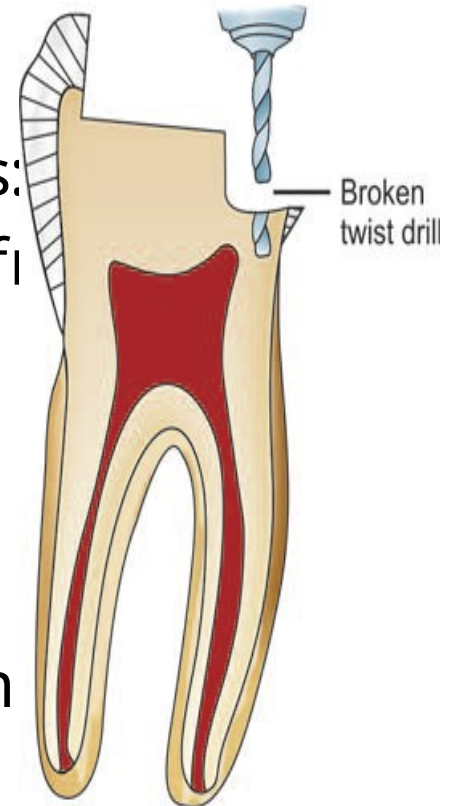


Fig.5.19: Broken twist drill

G. Loose Pins

- Loose pins can occur in following conditions:
 - Repeated insertion and removal of drill during pin preparation.
 - Pin drill is rotated more than required.
 - Pinhole is too large.
 - Manufacturer's discrepancy, i.e. poor quality control between pin drill and pin size.
- Pin failed to be driven in the pinhole resulting in stripped out or chipping of dentin or enamel.

To stabilize the pin, following can be done:

- Cement the existing pin in place.
- Drill another hole of the same diameter 1.5 mm away from the present hole and insert the same pin.

H. Pulpal Penetration and Periodontal Perforation

- Pin placement can also result in pulp and periodontal perforation (**Fig. 5.19**) and this is indicated with sudden bleeding while operating the drill. Bleeding is more abundant if there is periodontal perforation (**Fig.5.20**).

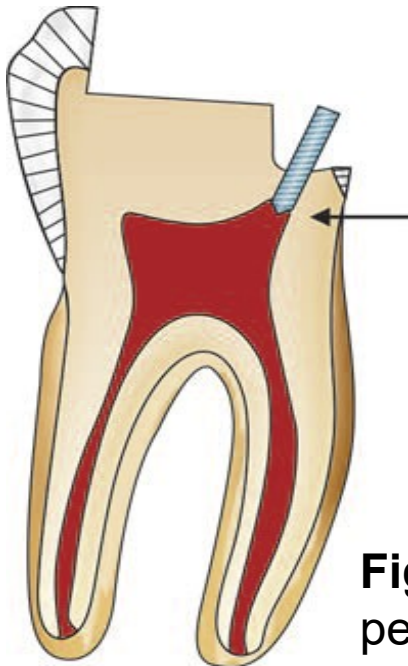


Fig. 5.19: Pulp perforation while pin placement

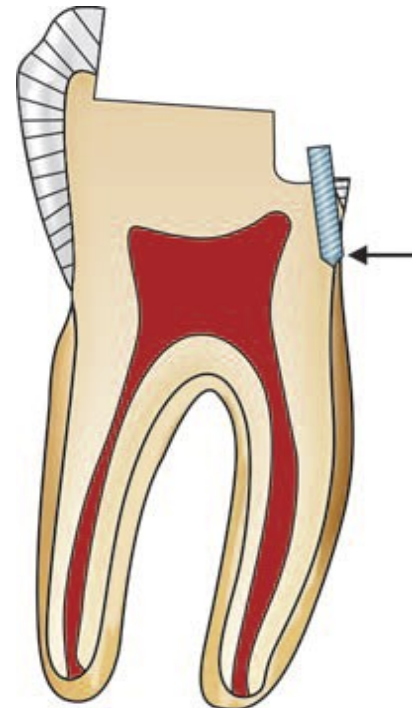


Fig.5,20: Perforation of periodontal space

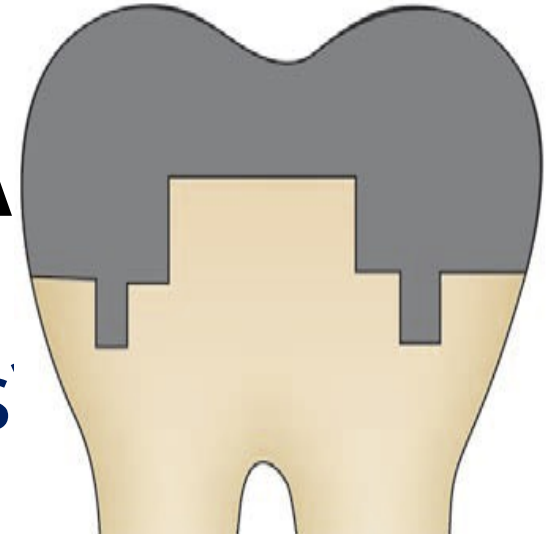
I. Heat Generation

- Generation of heat can be reduced by using 2.0 depth limiting drill and smallest possible pin.

J. Microleakage

- Microleakage around cemented pins occurs around whole circumference whereas it is semilunar in shape around threaded pins and friction locked pins.
- The dead space created by incomplete seating of pins may harbor bacteria and induce pulpal reaction.

AMA



What are Amalgapins Def.

- It is vertical posts of amalgam anchored in dentin. The pits prepared in dentin are shallow and little wider than pinholes and are called 'dentin chambers'.
- The post formed of the amalgam in the dentin chamber is called 'amalgapin'.
- They increase the retention and resistance. They also increase the bulk of amalgam. 'Dentin Chambers' for amalgapin techniques also increase the bulk of amalgam (Fig. 19.7)

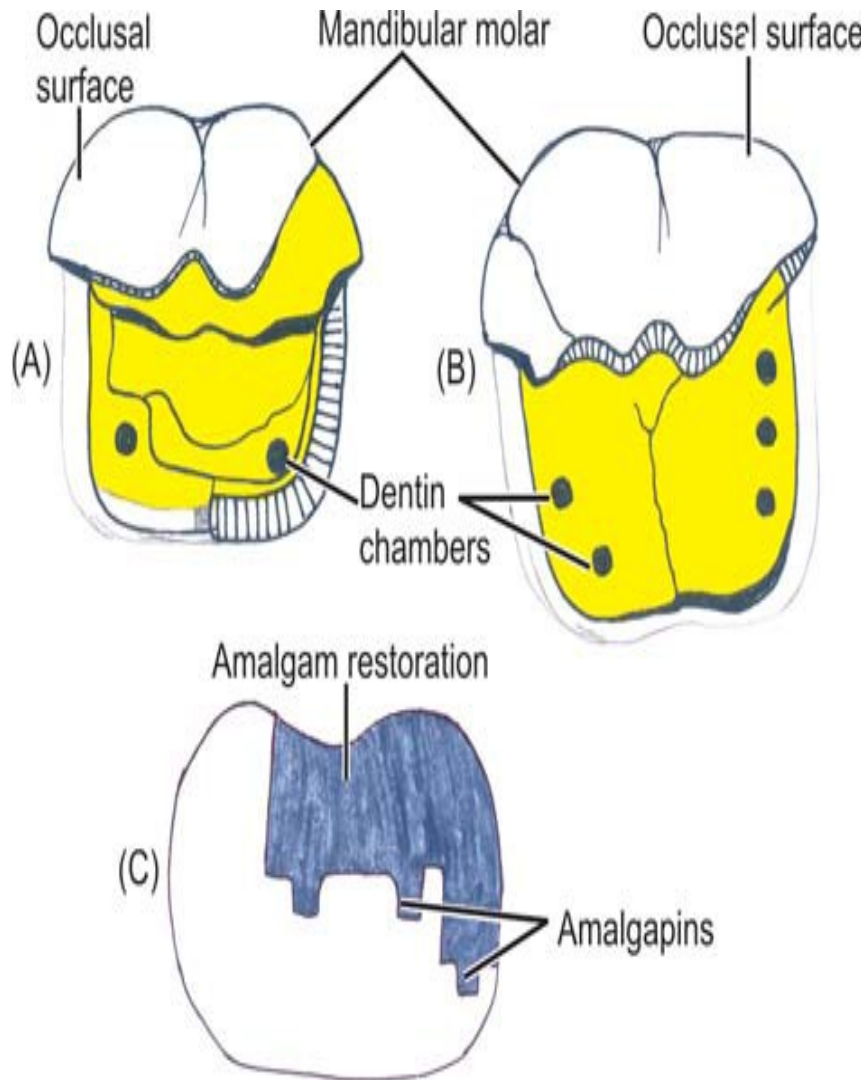


Fig.5. 30 Amalgapins technique

(A) and (B) are occlusal view of preparation of "dentin chambers" for amalgapin technique.

Positions of dentin chambers are relative to dentinoenamel junction and external tooth surface.

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